

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1 Product identifier

|              |                    |
|--------------|--------------------|
| Product name | Iloform RN 3017    |
| UFI:         | Q45-K0KD-Y00K-69W8 |
| Product code | 467692-DE15        |
| SDS #        | 467692             |
| Product type | Liquid.            |

### 1.2 Relevant identified uses of the substance or mixture and uses advised against

| Identified uses                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------|
| Use of lubricants in high energy open processes-Industrial<br>Use of lubricants in high energy open processes-Professional |

|                                  |                                                                                                                                           |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Use of the substance/<br>mixture | Metalworking fluid - neat.<br>For specific application advice see appropriate Technical Data Sheet or consult our company representative. |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|

### 1.3 Details of the supplier of the safety data sheet

|                |                                                                                                                                                                                                    |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Supplier       | Castrol Holdings Europe B.V.,<br>d'Arcyweg 76, 3198NA<br>Europoort<br>Rotterdam<br><br>Castrol Germany GmbH,<br>Überseeallee 1,<br>20457 Hamburg<br><br>+49 (0) 800 863 73 70<br>MSDSadvice@bp.com |
| E-mail address |                                                                                                                                                                                                    |

### 1.4 Emergency telephone number

|                               |                                       |
|-------------------------------|---------------------------------------|
| EMERGENCY<br>TELEPHONE NUMBER | Carechem: +44 (0) 1235 239 670 (24/7) |
|-------------------------------|---------------------------------------|

## SECTION 2: Hazards identification

### 2.1 Classification of the substance or mixture

|                                                                     |         |
|---------------------------------------------------------------------|---------|
| Product definition                                                  | Mixture |
| Classification according to Regulation (EC) No. 1272/2008 [CLP/GHS] |         |

Asp. Tox. 1, H304

Aquatic Chronic 3, H412

See Section 16 for the full text of the H statements declared above.

See sections 11 and 12 for more detailed information on health effects and symptoms and environmental hazards.

### 2.2 Label elements

|                   |                    |
|-------------------|--------------------|
| UFI:              | Q45-K0KD-Y00K-69W8 |
| Hazard pictograms |                    |



|             |        |
|-------------|--------|
| Signal word | Danger |
|-------------|--------|

|                   |                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------|
| Hazard statements | H304 - May be fatal if swallowed and enters airways.<br>H412 - Harmful to aquatic life with long lasting effects. |
|-------------------|-------------------------------------------------------------------------------------------------------------------|

### Precautionary statements

|            |                                          |
|------------|------------------------------------------|
| Prevention | P273 - Avoid release to the environment. |
|------------|------------------------------------------|

|                        |                 |               |                 |          |           |
|------------------------|-----------------|---------------|-----------------|----------|-----------|
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SECTION 2: Hazards identification

|                             |                                                                                                                                                       |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response                    | P301 + P310, P331 - IF SWALLOWED: Immediately call a POISON CENTER or physician. Do NOT induce vomiting.                                              |
| Storage                     | Not applicable.                                                                                                                                       |
| Disposal                    | P501 - Dispose of contents and container in accordance with all local, regional, national and international regulations.                              |
| Hazardous ingredients       | <div>Distillates (petroleum), hydrotreated light naphthenic</div> <div>Hydrocarbons, C16-C20, n-alkanes, isoalkanes, cyclics, aromatics (2-30%)</div> |
| Supplemental label elements | Not applicable.                                                                                                                                       |

EU Regulation (EC) No. 1907/2006 (REACH)

|                                                                                                                                    |                 |
|------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles | Not applicable. |
| Special packaging requirements                                                                                                     |                 |
| Containers to be fitted with child-resistant fastenings                                                                            | Not applicable. |
| Tactile warning of danger                                                                                                          | Not applicable. |

2.3 Other hazards

|                                                                                                            |                                                                                                                     |
|------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Product meets the criteria for PBT or vPvB according to Regulation (EC) No. 1907/2006, Annex XIII          | This mixture does not contain any substances that are assessed to be a PBT or a vPvB.                               |
| Product meets the criteria for endocrine disrupting properties according to Regulation (EC) No. 1907/2006. | This substance/mixture does not contain any components that are considered to have endocrine disrupting properties. |
| Other hazards which do not result in classification                                                        | Defatting to the skin.                                                                                              |

SECTION 3: Composition/information on ingredients

3.2 Mixtures

| Product definition                                                                                                                 |                                                                                      | Mixture   |                                                        |                                           |         |
|------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------|--------------------------------------------------------|-------------------------------------------|---------|
| Highly refined base oil (IP 346 DMSO extract < 3%). Proprietary performance additives.                                             |                                                                                      |           |                                                        |                                           |         |
| Product/ingredient name                                                                                                            | Identifiers                                                                          | %         | Classification                                         | Specific Conc. Limits, M-factors and ATEs | Type    |
| Distillates (petroleum), hydrotreated light naphthenic                                                                             | REACH #: 01-2119480375-34<br>EC: 265-156-6<br>CAS: 64742-53-6<br>Index: 649-466-00-2 | ≥50 - ≤75 | Asp. Tox. 1, H304                                      | -                                         | [1]     |
| Hydrocarbons, C16-C20, n-alkanes, isoalkanes, cyclics, aromatics (2-30%)                                                           | REACH #: 01-2119455996-19<br>EC: 919-006-8<br>CAS: -<br>Index: 649-223-00-0          | ≥10 - <20 | Asp. Tox. 1, H304<br>Aquatic Chronic 4, H413<br>EUH066 | -                                         | [1]     |
| Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%] | REACH #: 01-2119519251-50<br>EC: 700-990-0<br>CAS: -                                 | ≤3        | Aquatic Chronic 3, H412                                | -                                         | [1]     |
| 2,6-ditert-butyl-p-cresol                                                                                                          | REACH #: 01-2119555270-46<br>EC: 204-881-4<br>CAS: 128-37-0                          | ≤0.3      | Aquatic Acute 1, H400<br>Aquatic Chronic 1, H410       | M [Acute] = 1<br>M [Chronic] = 1          | [1] [2] |
| Reaction products of Dihydro-3-(tetrapropenyl)furan-2,5                                                                            | REACH #: 01-2120768184-49                                                            | ≤0.3      | Acute Tox. 4, H302<br>Skin Irrit. 2, H315              | ATE [Oral] = 500 mg/kg                    | [1]     |

|                        |                 |               |                 |          |         |
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**SECTION 3: Composition/information on ingredients**

dione with propane-1,2,diol

EC: 947-696-0

CAS: -

Eye Dam. 1, H318

Repr. 2, H361d

Aquatic Chronic 3, H412

**See Section 16 for the full text of the H statements declared above.**Type

[1] Substance classified with a health or environmental hazard

[2] Substance with a workplace exposure limit

Occupational exposure limits, if available, are listed in Section 8.

**SECTION 4: First aid measures****4.1 Description of first aid measures****Eye contact**

In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Eyelids should be held away from the eyeball to ensure thorough rinsing. Check for and remove any contact lenses. Get medical attention.

**Skin contact**

Wash skin thoroughly with soap and water or use recognised skin cleanser. Remove contaminated clothing and shoes. Wash clothing before reuse. Clean shoes thoroughly before reuse. Get medical attention if irritation develops.

**Inhalation**

If inhaled, remove to fresh air. Get medical attention if symptoms occur.

**Ingestion**

Do not induce vomiting. Never give anything by mouth to an unconscious person. If unconscious, place in recovery position and get medical attention immediately. Aspiration hazard if swallowed. Can enter lungs and cause damage. Get medical attention immediately.

**Protection of first-aiders**

No action shall be taken involving any personal risk or without suitable training. It may be dangerous to the person providing aid to give mouth-to-mouth resuscitation.

**4.2 Most important symptoms and effects, both acute and delayed**

See Section 11 for more detailed information on health effects and symptoms.

Potential acute health effects**Inhalation**

Vapour inhalation under ambient conditions is not normally a problem due to low vapour pressure.

**Ingestion**

Aspiration hazard if swallowed -- harmful or fatal if liquid is aspirated into lungs.

**Skin contact**

Defatting to the skin. May cause skin dryness and irritation.

**Eye contact**

See: Section 11. Toxicological Information - Potential acute health effects: Eye contact

Delayed and immediate effects as well as chronic effects from short and long-term exposure**Inhalation**

Overexposure to the inhalation of airborne droplets or aerosols may cause irritation of the respiratory tract.

**Ingestion**

Ingestion of large quantities may cause nausea and diarrhoea.

**Skin contact**

Prolonged or repeated contact can defat the skin and lead to irritation and/or dermatitis.

**Eye contact**

Potential risk of transient stinging or redness if accidental eye contact occurs.

**4.3 Indication of any immediate medical attention and special treatment needed****Notes to physician**

Treatment should in general be symptomatic and directed to relieving any effects. Product can be aspirated on swallowing or following regurgitation of stomach contents, and can cause severe and potentially fatal chemical pneumonitis, which will require urgent treatment. Because of the risk of aspiration, induction of vomiting and gastric lavage should be avoided. Gastric lavage should be undertaken only after endotracheal intubation. Monitor for cardiac dysrhythmias.

**SECTION 5: Firefighting measures****5.1 Extinguishing media****Suitable extinguishing media**

Use foam or all-purpose dry chemical to extinguish.

**Unsuitable extinguishing media**

Do not use water jet. The use of a water jet may cause the fire to spread by splashing the burning product.

**5.2 Special hazards arising from the substance or mixture****Hazards from the substance or mixture**

Swarf fires - Neat metal working oils may fume, thermally decompose or ignite if they come into contact with red hot swarf. To minimise the generation of red hot swarf ensure that a sufficient flow of oil is correctly directed to the cutting edge of the tool to flood it throughout cutting operations. As an additional precaution swarf should be regularly cleared from the immediate area to prevent the risk of fire. In a fire or if heated, a pressure increase will occur and the container may burst.

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## SECTION 5: Firefighting measures

|                                                       |                                                                                                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Hazardous combustion products</b>                  | Combustion products may include the following:<br>carbon oxides (CO, CO <sub>2</sub> ) (carbon monoxide, carbon dioxide)                                                                                                                                                                                                                                                     |
| <b>5.3 Advice for firefighters</b>                    |                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Special precautions for fire-fighters</b>          | No action shall be taken involving any personal risk or without suitable training. Promptly isolate the scene by removing all persons from the vicinity of the incident if there is a fire. This material is harmful to aquatic organisms. Fire water contaminated with this material must be contained and prevented from being discharged to any waterway, sewer or drain. |
| <b>Special protective equipment for fire-fighters</b> | Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode. Clothing for fire-fighters (including helmets, protective boots and gloves) conforming to European standard EN 469 will provide a basic level of protection for chemical incidents.                      |

## SECTION 6: Accidental release measures

|                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>6.1 Personal precautions, protective equipment and emergency procedures</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>For non-emergency personnel</b>                                             | Contact emergency personnel. No action shall be taken involving any personal risk or without suitable training. Evacuate surrounding areas. Keep unnecessary and unprotected personnel from entering. Do not touch or walk through spilt material. Floors may be slippery; use care to avoid falling. Avoid breathing vapour or mist. Provide adequate ventilation. Put on appropriate personal protective equipment.                                                                                  |
| <b>For emergency responders</b>                                                | Entry into a confined space or poorly ventilated area contaminated with vapour, mist or fume is extremely hazardous without the correct respiratory protective equipment and a safe system of work. Wear self-contained breathing apparatus. Wear a suitable chemical protective suit. Chemical resistant boots. See also the information in "For non-emergency personnel".                                                                                                                            |
| <b>6.2 Environmental precautions</b>                                           | Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air). Water polluting material. May be harmful to the environment if released in large quantities.                                                                                                                                                                                     |
| <b>6.3 Methods and material for containment and cleaning up</b>                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Small spill</b>                                                             | Stop leak if without risk. Move containers from spill area. Absorb with an inert material and place in an appropriate waste disposal container. Dispose of via a licensed waste disposal contractor.                                                                                                                                                                                                                                                                                                   |
| <b>Large spill</b>                                                             | Stop leak if without risk. Move containers from spill area. Approach the release from upwind. Prevent entry into sewers, water courses, basements or confined areas. Contain and collect spillage with non-combustible, absorbent material e.g. sand, earth, vermiculite or diatomaceous earth and place in container for disposal according to local regulations. Contaminated absorbent material may pose the same hazard as the spilt product. Dispose of via a licensed waste disposal contractor. |
| <b>6.4 Reference to other sections</b>                                         | See Section 1 for emergency contact information.<br>See Section 5 for firefighting measures.<br>See Section 8 for information on appropriate personal protective equipment.<br>See Section 12 for environmental precautions.<br>See Section 13 for additional waste treatment information.                                                                                                                                                                                                             |

## SECTION 7: Handling and storage

The information in this section contains generic advice and guidance. The list of Identified Uses in Section 1 should be consulted for any available use-specific information provided in the Exposure Scenario(s).

|                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>7.1 Precautions for safe handling</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Protective measures</b>               | Put on appropriate personal protective equipment. Do not swallow. Aspiration hazard if swallowed. Can enter lungs and cause damage. Never siphon by mouth. Avoid contact with eyes, skin and clothing. Avoid breathing vapour or mist. Avoid contact of spilt material and runoff with soil and surface waterways. Keep in the original container or an approved alternative made from a compatible material, kept tightly closed when not in use. Do not reuse container. Empty containers retain product residue and can be hazardous. Concentrations of mist, fumes and vapours in enclosed spaces may result in the formation of explosive atmospheres. Excessive splashing, agitation or heating must be avoided. During metal working, solid particles from workpieces or tools will contaminate the fluid and may cause abrasions of the skin. Where such abrasions result in a penetration of the skin, first aid treatment should be applied as soon as reasonably possible. The presence of certain metals in the workpiece or tool, such as chromium, cobalt and nickel, can contaminate the metalworking fluid, as can bacteria, and as a result may induce allergic and other skin reactions, especially if personal hygiene is inadequate. |

|                               |                 |                      |                 |                 |         |
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SECTION 7: Handling and storage

|                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Advice on general occupational hygiene                           | Eating, drinking and smoking should be prohibited in areas where this material is handled, stored and processed. Wash thoroughly after handling. Remove contaminated clothing and protective equipment before entering eating areas. See also Section 8 for additional information on hygiene measures.                                                                                                                                                                                                                                                        |
| 7.2 Conditions for safe storage, including any incompatibilities | Store in accordance with local regulations. Store in a dry, cool and well-ventilated area, away from incompatible materials (see Section 10). Store locked up. Keep away from heat and direct sunlight. Keep container tightly closed and sealed until ready for use. Containers that have been opened must be carefully resealed and kept upright to prevent leakage. Store and use only in equipment/containers designed for use with this product. Do not store in unlabelled containers. Use appropriate containment to avoid environmental contamination. |
| Not suitable                                                     | Prolonged exposure to elevated temperature.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Germany - Storage code                                           | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| 7.3 Specific end use(s)                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Recommendations                                                  | See section 1.2 and Exposure scenarios in annex, if applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

SECTION 8: Exposure controls/personal protection

The information in this section contains generic advice and guidance. The list of Identified Uses in Section 1 should be consulted for any available use-specific information provided in the Exposure Scenario(s).

8.1 Control parameters

Occupational exposure limits

| Product/ingredient name   | Exposure limit values                                                                                                                                                                     |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2,6-ditert-butyl-p-cresol | <b>TRGS 900 OEL (Germany)</b><br>TWA 8 hours: 10 mg/m³. Form: Inhalable fraction. Issued/Revised: 7/2013.<br>PEAK 15 minutes: 40 mg/m³. Form: Inhalable fraction. Issued/Revised: 7/2013. |

Whilst specific OELs for certain components may be shown in this section, other components may be present in any mist, vapour or dust produced. Therefore, the specific OELs may not be applicable to the product as a whole and are provided for guidance only.

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Recommended monitoring procedures | Reference should be made to monitoring standards, such as the following: European Standard EN 689 (Workplace atmospheres - Guidance for the assessment of exposure by inhalation to chemical agents for comparison with limit values and measurement strategy) European Standard EN 14042 (Workplace atmospheres - Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents) European Standard EN 482 (Workplace atmospheres - General requirements for the performance of procedures for the measurement of chemical agents) Reference to national guidance documents for methods for the determination of hazardous substances will also be required. |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Biological exposure indices

| Product/ingredient name   | Exposure indices                                                                                                                                                             |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2,6-ditert-butyl-p-cresol | <b>DFG BEI-values list (Germany, 7/2024)</b><br>BEI: 7 µg/l, Butylated hydroxytoluene acid (after hydrolysis) [in urine].<br>Sampling time: end of exposure or end of shift. |

DNELs/DMELs

| Product/ingredient name    | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2,6-di-tert-butyl-p-cresol | <b>DNEL - Workers - Long term - Inhalation</b><br>1.76 mg/m³<br>Effects: Systemic<br><br><b>DNEL - Workers - Long term - Dermal</b><br>0.5 mg/kg bw/day<br>Effects: Systemic<br><br><b>DNEL - General population - Long term - Inhalation</b><br>0.435 mg/m³<br>Effects: Systemic<br><br><b>DNEL - General population - Long term - Dermal</b><br>0.25 mg/kg bw/day<br>Effects: Systemic<br><br><b>DNEL - General population - Long term - Oral</b><br>0.25 mg/kg bw/day |

|                        |                 |               |                 |          |         |
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SECTION 8: Exposure controls/personal protection

Effects: Systemic

PNECs

Product/ingredient name

2,6-di-tert-butyl-p-cresol

Result

Fresh water

0.199 µg/l

Marine water

0.02 µg/l

Sewage Treatment Plant

0.17 mg/l

Fresh water sediment

0.458 mg/l

Marine water sediment

0.046 mg/l

Soil

0.054 mg/l

8.2 Exposure controls

Appropriate engineering controls

Provide exhaust ventilation or other engineering controls to keep the relevant airborne concentrations below their respective occupational exposure limits. All activities involving chemicals should be assessed for their risks to health, to ensure exposures are adequately controlled. Personal protective equipment should only be considered after other forms of control measures (e.g. engineering controls) have been suitably evaluated. Personal protective equipment should conform to appropriate standards, be suitable for use, be kept in good condition and properly maintained. Your supplier of personal protective equipment should be consulted for advice on selection and appropriate standards. For further information contact your national organisation for standards. The final choice of protective equipment will depend upon a risk assessment. It is important to ensure that all items of personal protective equipment are compatible.

Individual protection measures

Hygiene measures

Wash hands, forearms and face thoroughly after handling chemical products, before eating, smoking and using the lavatory and at the end of the working period. Ensure that eyewash stations and safety showers are close to the workstation location.

Respiratory protection

In case of insufficient ventilation, wear suitable respiratory equipment. For protection against metal working fluids, respiratory protection that is classified as “resistant to oil” (class R) or oil proof (class P) should be selected where appropriate. Depending on the level of airborne contaminants, an air-purifying, half-mask respirator (with HEPA filter) including disposable (P- or R-series) (for oil mists less than 50mg/m3), or any powered, air-purifying respirator equipped with hood or helmet and HEPA filter (for oil mists less than 125 mg/m3). Where organic vapours are a potential hazard during metalworking operations, a combination particulate and organic vapour filter may be necessary. The correct choice of respiratory protection depends upon the chemicals being handled, the conditions of work and use, and the condition of the respiratory equipment. Safety procedures should be developed for each intended application. Respiratory protection equipment should therefore be chosen in consultation with the supplier/manufacturer and with a full assessment of the working conditions.

Eye/face protection

Safety glasses with side shields.

Skin protection

Hand protection

General Information:

Because specific work environments and material handling practices vary, safety procedures should be developed for each intended application. The correct choice of protective gloves depends upon the chemicals being handled, and the conditions of work and use. Most gloves provide protection for only a limited time before they must be discarded and replaced (even the best chemically resistant gloves will break down after repeated chemical exposures).

Gloves should be chosen in consultation with the supplier / manufacturer and taking account of a full assessment of the working conditions.

Recommended: Nitrile gloves.

Breakthrough time:

Breakthrough time data are generated by glove manufacturers under laboratory test conditions

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and represent how long a glove can be expected to provide effective permeation resistance. It is important when following breakthrough time recommendations that actual workplace conditions are taken into account. Always consult with your glove supplier for up-to-date technical information on breakthrough times for the recommended glove type. Our recommendations on the selection of gloves are as follows:

Continuous contact:

Gloves with a minimum breakthrough time of 240 minutes, or >480 minutes if suitable gloves can be obtained.  
If suitable gloves are not available to offer that level of protection, gloves with shorter breakthrough times may be acceptable as long as appropriate glove maintenance and replacement regimes are determined and adhered to.

Short-term / splash protection:

Recommended breakthrough times as above.  
It is recognised that for short-term, transient exposures, gloves with shorter breakthrough times may commonly be used. Therefore, appropriate maintenance and replacement regimes must be determined and rigorously followed.

Glove Thickness:

For general applications, we recommend gloves with a thickness typically greater than 0.35 mm.

It should be emphasised that glove thickness is not necessarily a good predictor of glove resistance to a specific chemical, as the permeation efficiency of the glove will be dependent on the exact composition of the glove material. Therefore, glove selection should also be based on consideration of the task requirements and knowledge of breakthrough times. Glove thickness may also vary depending on the glove manufacturer, the glove type and the glove model. Therefore, the manufacturers' technical data should always be taken into account to ensure selection of the most appropriate glove for the task.

Note: Depending on the activity being conducted, gloves of varying thickness may be required for specific tasks. For example:

- Thinner gloves (down to 0.1 mm or less) may be required where a high degree of manual dexterity is needed. However, these gloves are only likely to give short duration protection and would normally be just for single use applications, then disposed of.
- Thicker gloves (up to 3 mm or more) may be required where there is a mechanical (as well as a chemical) risk i.e. where there is abrasion or puncture potential.

Skin and body

Use of protective clothing is good industrial practice.  
Personal protective equipment for the body should be selected based on the task being performed and the risks involved and should be approved by a specialist before handling this product.  
Cotton or polyester/cotton overalls will only provide protection against light superficial contamination that will not soak through to the skin. Overalls should be laundered on a regular basis. When the risk of skin exposure is high (e.g. when cleaning up spillages or if there is a risk of splashing) then chemical resistant aprons and/or impervious chemical suits and boots will be required.

Refer to standards:

Respiratory protection: EN 529  
Gloves: EN 420, EN 374  
Eye protection: EN 166  
Filtering half-mask: EN 149  
Filtering half-mask with valve: EN 405  
Half-mask: EN 140 plus filter  
Full-face mask: EN 136 plus filter  
Particulate filters: EN 143  
Gas/combined filters: EN 14387

Environmental exposure controls

Emissions from ventilation or work process equipment should be checked to ensure they comply with the requirements of environmental protection legislation. In some cases, fume scrubbers, filters or engineering modifications to the process equipment will be necessary to reduce emissions to acceptable levels.

SECTION 9: Physical and chemical properties

The conditions of measurement of all properties are at standard temperature and pressure unless otherwise indicated.

9.1 Information on basic physical and chemical properties

| Physical state                                         | Liquid.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-------------------------|-------|-------------------------|-------------------------------------------------|-----|-------|-----|--------|-------|-----|--------|--------------------------------------------------------|----------|-------|-------------|--|--|--|---------------------------------------|------------|-------------|--|--|--|--|
| Colour                                                 | Colourless.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Odour                                                  | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Odour threshold                                        | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Melting point/freezing point                           | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Initial boiling point and boiling range                | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Flammability                                           | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Lower and upper explosion limit                        | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Flash point                                            | Open cup: >125°C (>257°F) [Cleveland DIN EN ISO 2592]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Auto-ignition temperature                              | <table><tr><th>Ingredient name</th><th>°C</th><th>°F</th><th>Method</th></tr><tr><td>Distillates (petroleum), hydrosulfurized middle</td><td>225</td><td>437</td><td></td></tr></table>                                                                                                                                                                                                                                                                                                                                                       | Ingredient name | °C                      | °F    | Method                  | Distillates (petroleum), hydrosulfurized middle | 225 | 437   |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Ingredient name                                        | °C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | °F              | Method                  |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Distillates (petroleum), hydrosulfurized middle        | 225                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 437             |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Decomposition temperature                              | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| pH                                                     | Not applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Kinematic viscosity                                    | Kinematic: 7.5 mm²/s (7.5 cSt) at 40°C<br>Kinematic: 2.1 mm²/s (2.1 cSt) at 100°C                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Solubility                                             | <table><tr><th>Media</th><th>Result</th></tr><tr><td>water</td><td>Not soluble</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                              | Media           | Result                  | water | Not soluble             |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Media                                                  | Result                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| water                                                  | Not soluble                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Partition coefficient n-octanol/ water (log value)     | Not applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Vapour pressure                                        | <table><tr><th rowspan="2">Ingredient name</th><th colspan="2">Vapour Pressure at 20°C</th><th colspan="3">Vapour pressure at 50°C</th></tr><tr><th>mm Hg</th><th>kPa</th><th>Method</th><th>mm Hg</th><th>kPa</th><th>Method</th></tr><tr><td>Distillates (petroleum), hydrotreated heavy naphthenic</td><td>&lt;0.07501</td><td>&lt;0.01</td><td>ASTM D 5191</td><td></td><td></td><td></td></tr><tr><td>Phenol, isobutylated, phosphate (3:1)</td><td>0.00000031</td><td>0.000000041</td><td></td><td></td><td></td><td></td></tr></table> | Ingredient name | Vapour Pressure at 20°C |       | Vapour pressure at 50°C |                                                 |     | mm Hg | kPa | Method | mm Hg | kPa | Method | Distillates (petroleum), hydrotreated heavy naphthenic | <0.07501 | <0.01 | ASTM D 5191 |  |  |  | Phenol, isobutylated, phosphate (3:1) | 0.00000031 | 0.000000041 |  |  |  |  |
| Ingredient name                                        | Vapour Pressure at 20°C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                 | Vapour pressure at 50°C |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
|                                                        | mm Hg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | kPa             | Method                  | mm Hg | kPa                     | Method                                          |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Distillates (petroleum), hydrotreated heavy naphthenic | <0.07501                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <0.01           | ASTM D 5191             |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Phenol, isobutylated, phosphate (3:1)                  | 0.00000031                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.000000041     |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Density and/or Relative density                        | <1000 kg/m³ (<1 g/cm³) at 15°C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Relative vapour density                                | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Particle characteristics                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Median particle size                                   | Not applicable.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| 9.2 Other information                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Evaporation rate                                       | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Explosive properties                                   | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |
| Oxidising properties                                   | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                         |       |                         |                                                 |     |       |     |        |       |     |        |                                                        |          |       |             |  |  |  |                                       |            |             |  |  |  |  |

SECTION 10: Stability and reactivity

|                                         |                                                                                                                                                                         |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.1 Reactivity                         | No specific test data available for this product. Refer to Conditions to avoid and Incompatible materials for additional information.                                   |
| 10.2 Chemical stability                 | The product is stable.                                                                                                                                                  |
| 10.3 Possibility of hazardous reactions | Under normal conditions of storage and use, hazardous reactions will not occur.<br>Under normal conditions of storage and use, hazardous polymerisation will not occur. |
| 10.4 Conditions to avoid                | Avoid all possible sources of ignition (spark or flame).                                                                                                                |
| 10.5 Incompatible materials             | Reactive or incompatible with the following materials: oxidising materials.                                                                                             |

SECTION 10: Stability and reactivity

10.6 Hazardous decomposition products Under normal conditions of storage and use, hazardous decomposition products should not be produced.

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Acute toxicity

Product/ingredient name

Distillates (petroleum), hydrotreated light naphthenic

Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%]

2,6-di-tert-butyl-p-cresol

Result

Rat - Oral - LD50  
>5000 mg/kg  
OECD 401

Rabbit - Dermal - LD50  
>5000 mg/kg  
OECD 402

Rat - Inhalation - LC50 Dusts and mists  
>5.53 mg/l [4 hours]  
OECD 403

Rat - Oral - LD50  
>5000 mg/kg  
OECD 401

Rabbit - Dermal - LD50  
>2000 mg/kg  
OECD 402

Rat - Inhalation - LC50 Dusts and mists  
>0.4 mg/l [6 hours]

Rat - Oral - LD50  
>5000 mg/kg  
OECD 401

Rat - Dermal - LD50  
>2000 mg/kg  
OECD 402

Acute toxicity estimates

| Product/ingredient name                                                              | Oral (mg/kg) | Dermal (mg/kg) | Inhalation (gases) (ppm) | Inhalation (vapours) (mg/l) | Inhalation (dusts and mists) (mg/l) |
|--------------------------------------------------------------------------------------|--------------|----------------|--------------------------|-----------------------------|-------------------------------------|
| Reaction products of Dihydro-3-(tetrapropenyl) furan-2,5 dione with propane-1,2,diol | 500          | N/A            | N/A                      | N/A                         | N/A                                 |

Skin corrosion/irritation

Product/ingredient name

Distillates (petroleum), hydrotreated light naphthenic

Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%]

2,6-di-tert-butyl-p-cresol

Result

Rabbit - Skin - Non-irritant to skin.

Rabbit - Skin - Not irritant  
OECD 404

Rabbit - Skin - Non-irritant to skin.  
OECD 404

Serious eye damage/eye irritation

Product/ingredient name

Result

SECTION 11: Toxicological information

Distillates (petroleum), hydrotreated light naphthenic

Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%]

2,6-di-tert-butyl-p-cresol

Rabbit - Eyes - Non-irritating to the eyes. OECD 405

Rabbit - Eyes - Not irritant OECD 405

Rabbit - Eyes - Non-irritating to the eyes. OECD 405

Respiratory corrosion/irritation

Not available.

Respiratory or skin sensitization

Product/ingredient name

Distillates (petroleum), hydrotreated light naphthenic

Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%]

2,6-di-tert-butyl-p-cresol

Result

Guinea pig - skin  
OECD 406  
Result: Not sensitising

Mouse - skin  
OECD 429  
Result: Sensitising

Guinea pig - skin  
OECD 406  
Result: Not sensitising

Germ cell mutagenicity

Product/ingredient name

Distillates (petroleum), hydrotreated light naphthenic

Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%]

2,6-di-tert-butyl-p-cresol

Result

In vitro - Bacteria  
OECD 471 [Bacterial Reverse Mutation Test]  
Result: Negative

In vitro - Mammalian-Animal  
OECD 473 [In vitro Mammalian Chromosomal Aberration Test]  
Result: Negative

In vitro - Mammalian-Animal  
In vitro Mammalian Cell Gene Mutation Test  
Result: Equivocal

In vitro - Bacteria  
OECD 471  
Result: Negative

In vitro - Mammalian-Animal  
OECD 479  
Result: Negative

In vitro - Mammalian-Animal  
OECD 476  
Result: Negative

In vitro - Bacteria  
Bacterial Reverse Mutation Test  
Result: Negative

In vivo - Mammal - species unspecified  
Mammalian Erythrocyte Micronucleus Test  
Result: Negative

**SECTION 11: Toxicological information****Carcinogenicity****Product/ingredient name**

☒ Distillates (petroleum), hydrotreated light naphthenic

**Result**

**Mouse - Dermal - Unspecified**

OECD 451

Result: Negative

2,6-di-tert-butyl-p-cresol

**Rat - Oral - Unspecified**

Result: Negative

**Reproductive toxicity****Product/ingredient name**

☒ Distillates (petroleum), hydrotreated light naphthenic

**Result**

**Rat - Oral**

OECD 421

Maternal toxicity: Negative

Fertility effects: Negative

Developmental: Negative

Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%]

**Rat - Oral**

OECD 421

Maternal toxicity: Negative

Fertility effects: Negative

Developmental: Negative

2,6-di-tert-butyl-p-cresol

**Rat - Oral**

OECD 416

Maternal toxicity: Positive

Fertility effects: Negative

Developmental: Negative

**Specific target organ toxicity (single exposure)**

Not available.

**Specific target organ toxicity (repeated exposure)**

Not available.

**Aspiration hazard****Product/ingredient name**

☒ Distillates (petroleum), hydrotreated light naphthenic

**Result**

ASPIRATION HAZARD - Category 1

Hydrocarbons, C16-C20, n-alkanes, isoalkanes, cyclics, aromatics (2-30%)

ASPIRATION HAZARD - Category 1

**Information on likely routes of exposure**

Routes of entry anticipated: Oral, Dermal, Inhalation, Eyes.

**Potential acute health effects****Inhalation**

Vapour inhalation under ambient conditions is not normally a problem due to low vapour pressure.

**Ingestion**

Aspiration hazard if swallowed -- harmful or fatal if liquid is aspirated into lungs.

**Skin contact**

Defatting to the skin. May cause skin dryness and irritation.

**Eye contact**

☒ No known significant effects or critical hazards.

**Symptoms related to the physical, chemical and toxicological characteristics****Inhalation**

No specific data.

**Ingestion**

Adverse symptoms may include the following:  
nausea or vomiting

**Skin contact**

Adverse symptoms may include the following:  
irritation  
dryness  
cracking

**Eye contact**

No specific data.

**Delayed and immediate effects as well as chronic effects from short and long-term exposure****Inhalation**

Overexposure to the inhalation of airborne droplets or aerosols may cause irritation of the respiratory tract.

**Product name** Iloform RN 3017

**Product code** 467692-DE15

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(Germany)

SECTION 11: Toxicological information

|                                         |                                                                                            |
|-----------------------------------------|--------------------------------------------------------------------------------------------|
| Ingestion                               | Ingestion of large quantities may cause nausea and diarrhoea.                              |
| Skin contact                            | Prolonged or repeated contact can defat the skin and lead to irritation and/or dermatitis. |
| Eye contact                             | Potential risk of transient stinging or redness if accidental eye contact occurs.          |
| <u>Potential chronic health effects</u> |                                                                                            |
| Not available.                          |                                                                                            |
| Conclusion/Summary [Product]            | Not available.                                                                             |
| General                                 | No known significant effects or critical hazards.                                          |
| Carcinogenicity                         | No known significant effects or critical hazards.                                          |
| Mutagenicity                            | No known significant effects or critical hazards.                                          |
| Developmental effects                   | No known significant effects or critical hazards.                                          |
| Fertility effects                       | No known significant effects or critical hazards.                                          |

11.2 Information on other hazards

11.2.1 Endocrine disrupting properties

|                              |                                                                                                                                                                                                       |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conclusion/Summary [Product] |  This substance/mixture does not contain any components that are considered to have endocrine disrupting properties. |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

11.2.2 Other information

Not available.

SECTION 12: Ecological information

12.1 Toxicity

| Product/ingredient name                                                                                                                  | Result                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|  Distillates (petroleum), hydrotreated light naphthenic | <b>Acute - EL50</b><br>OECD 201<br>Algae<br>>100 mg/l [72 hours]     |
|                                                                                                                                          | <b>Acute - EL50</b><br>OECD 202<br>Daphnia<br>>10000 mg/l [48 hours] |
|                                                                                                                                          | <b>Acute - LL50</b><br>OECD 203<br>Fish<br>>100 mg/l [96 hours]      |
|                                                                                                                                          | <b>Chronic - NOEL</b><br>OECD 201<br>Algae<br>≥100 mg/l [72 hours]   |
|                                                                                                                                          | <b>Chronic - NOEL</b><br>OECD 211<br>Daphnia<br>10 mg/l [21 days]    |
|                                                                                                                                          | <b>Acute - EC50</b><br>OECD 201<br>Algae<br>1.4 mg/l [72 hours]      |
|                                                                                                                                          | <b>Acute - EC50</b><br>OECD 202<br>Daphnia<br>202 mg/l [48 hours]    |
|                                                                                                                                          | <b>Acute - LC50</b><br>OECD 203<br>Fish<br>0.8 mg/l [96 hours]       |
|                                                                                                                                          | <b>Chronic - NOEC</b><br>OECD 201<br>Algae<br>0.05 mg/l [72 hours]   |
|                                                                                                                                          |                                                                      |
| Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%]       |                                                                      |

SECTION 12: Ecological information

|                            |                                                                      |
|----------------------------|----------------------------------------------------------------------|
| 2,6-di-tert-butyl-p-cresol | <b>Chronic - NOEC</b><br>OECD 211<br>Daphnia<br>0.032 mg/l [21 days] |
|                            | <b>Acute - EC50</b><br>OECD 201<br>Algae<br>>0.4 mg/l [72 hours]     |
|                            | <b>Acute - EC50</b><br>OECD 202<br>Daphnia<br>0.48 mg/l [48 hours]   |
|                            | <b>Acute - LC50</b><br>OECD 203<br>Fish<br>>0.57 mg/l [96 hours]     |
|                            | <b>Chronic - NOEC</b><br>OECD 201<br>Algae<br>0.4 mg/l [72 hours]    |
|                            | <b>Chronic - NOEC</b><br>OECD 211<br>Daphnia<br>0.069 mg/l [21 days] |
|                            | <b>Chronic - NOEC</b><br>OECD 210<br>Fish<br>0.053 mg/l [30 days]    |
|                            |                                                                      |
|                            |                                                                      |
|                            |                                                                      |

**Environmental hazards** Harmful to aquatic life with long lasting effects.

12.2 Persistence and degradability

Expected to be biodegradable.

| Product/ingredient name                                | Result                                    |
|--------------------------------------------------------|-------------------------------------------|
| Distillates (petroleum), hydrotreated light naphthenic | OECD 301F<br>31% [28 days] - Not readily  |
| 2,6-di-tert-butyl-p-cresol                             | OECD 301C<br>4.5% [28 days] - Not readily |

12.3 Bioaccumulative potential

Not available.

| Product/ingredient name                                                                                                                  | LogP <sub>ow</sub> | BCF         | Potential |
|------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------|-----------|
| Reaction mass of p-tert-butylphenyldiphenyl phosphate and bis(p-tert-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%] | 4.85               | -           | High      |
| 2,6-di-tert-butyl-p-cresol                                                                                                               | 5.1                | 330 to 1800 | High      |

12.4 Mobility in soil

Soil/water partition coefficient

| Product/ingredient name    | logK <sub>oc</sub> | K <sub>oc</sub> |
|----------------------------|--------------------|-----------------|
| 2,6-di-tert-butyl-p-cresol | 3.65               | 4489.84         |

Results of PMT and vPvM assessment

SECTION 12: Ecological information

| Product/ingredient name                                                                                                            | PMT | P  | M  | T  | vPvM | vP | vM |
|------------------------------------------------------------------------------------------------------------------------------------|-----|----|----|----|------|----|----|
| Distillates (petroleum), hydrotreated light naphthenic                                                                             | No  | No | No | No | No   | No | No |
| Hydrocarbons, C16-C20, n-alkanes, isoalkanes, cyclics, aromatics (2-30%)                                                           | No  | No | No | No | No   | No | No |
| Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%] | No  | No | No | No | No   | No | No |
| 2,6-di-tert-butyl-p-cresol                                                                                                         | No  | No | No | No | No   | No | No |
| Reaction products of Dihydro-3-(tetrapropenyl) furan-2,5 dione with propane-1,2,diol                                               | No  | No | No | No | No   | No | No |

Mobility Liquid. insoluble in water.

Conclusion/Summary The product does not meet the criteria to be considered as a PMT or vPvM.

12.5 Results of PBT and vPvB assessment

Regulation (EC) No. 1907/2006 [REACH]

| Product/ingredient name                                                                                                            | PBT | P   | B   | T   | vPvB | vP  | vB  |
|------------------------------------------------------------------------------------------------------------------------------------|-----|-----|-----|-----|------|-----|-----|
| Distillates (petroleum), hydrotreated light naphthenic                                                                             | No  | N/A | N/A | No  | N/A  | N/A | N/A |
| Hydrocarbons, C16-C20, n-alkanes, isoalkanes, cyclics, aromatics (2-30%)                                                           | No  | N/A | N/A | No  | N/A  | N/A | N/A |
| Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%] | No  | N/A | N/A | No  | N/A  | N/A | N/A |
| 2,6-di-tert-butyl-p-cresol                                                                                                         | No  | N/A | No  | No  | No   | N/A | No  |
| Reaction products of Dihydro-3-(tetrapropenyl) furan-2,5 dione with propane-1,2,diol                                               | N/A | N/A | N/A | Yes | N/A  | N/A | N/A |

Regulation (EC) No. 1272/2008 [CLP]

| Product/ingredient name                                                                                                            | PBT | P  | B  | T  | vPvB | vP | vB |
|------------------------------------------------------------------------------------------------------------------------------------|-----|----|----|----|------|----|----|
| Distillates (petroleum), hydrotreated light naphthenic                                                                             | No  | No | No | No | No   | No | No |
| Hydrocarbons, C16-C20, n-alkanes, isoalkanes, cyclics, aromatics (2-30%)                                                           | No  | No | No | No | No   | No | No |
| Reaction mass of p-t-butylphenyldiphenyl phosphate and bis(p-t-butylphenyl)phenyl phosphate and triphenyl phosphate [TPP>2,5 <25%] | No  | No | No | No | No   | No | No |
| 2,6-di-tert-butyl-p-cresol                                                                                                         | No  | No | No | No | No   | No | No |
| Reaction products of Dihydro-3-(tetrapropenyl) furan-2,5 dione with propane-1,2,diol                                               | No  | No | No | No | No   | No | No |

Conclusion/Summary The product does not meet the criteria to be considered as a PBT or vPvB.

Regulation (EC) No. 1272/2008 [CLP]

12.6 Endocrine disrupting properties

Conclusion/Summary [Product] This substance/mixture does not contain any components that are considered to have endocrine disrupting properties.

12.7 Other adverse effects No known significant effects or critical hazards.

|                        |                 |               |                 |          |         |
|------------------------|-----------------|---------------|-----------------|----------|---------|
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**SECTION 13: Disposal considerations**

The information in this section contains generic advice and guidance. The list of Identified Uses in Section 1 should be consulted for any available use-specific information provided in the Exposure Scenario(s).

**13.1 Waste treatment methods****Product****Methods of disposal**

Where possible, arrange for product to be recycled. Dispose of via an authorised person/ licensed waste disposal contractor in accordance with local regulations.

**Hazardous waste**

Yes.

**European waste catalogue (EWC)**

| Waste code | Waste designation                                                              |
|------------|--------------------------------------------------------------------------------|
| 12 01 07*  | mineral-based machining oils free of halogens (except emulsions and solutions) |

However, deviation from the intended use and/or the presence of any potential contaminants may require an alternative waste disposal code to be assigned by the end user.

**Packaging****Methods of disposal**

Where possible, arrange for product to be recycled. Dispose of via an authorised person/ licensed waste disposal contractor in accordance with local regulations.

| Waste code | European waste catalogue (EWC)                                           |
|------------|--------------------------------------------------------------------------|
| 15 01 10*  | packaging containing residues of or contaminated by hazardous substances |

**Special precautions**

This material and its container must be disposed of in a safe way. Care should be taken when handling emptied containers that have not been cleaned or rinsed out. Empty containers or liners may retain some product residues. Empty containers represent a fire hazard as they may contain flammable product residues and vapour. Never weld, solder or braze empty containers. Avoid dispersal of spilt material and runoff and contact with soil, waterways, drains and sewers.

**References**

Commission 2014/955/EU  
Directive 2008/98/EC

**SECTION 14: Transport information**

|                                        | ADR/RID        | ADN            | IMDG           | IATA           |
|----------------------------------------|----------------|----------------|----------------|----------------|
| <b>14.1 UN number or ID number</b>     | Not regulated. | Not regulated. | Not regulated. | Not regulated. |
| <b>14.2 UN proper shipping name</b>    | -              | -              | -              | -              |
| <b>14.3 Transport hazard class(es)</b> | -              | -              | -              | -              |
| <b>14.4 Packing group</b>              | -              | -              | -              | -              |
| <b>14.5 Environmental hazards</b>      | No.            | No.            | No.            | No.            |
| <b>Additional information</b>          | -              | -              | -              | -              |

**14.6 Special precautions for user**

Not available.

**14.7 Maritime transport in bulk according to IMO instruments**

Not available.

**SECTION 15: Regulatory information****15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture****EU Regulation (EC) No. 1907/2006 (REACH)****Annex XIV - List of substances subject to authorisation****Annex XIV**

None of the components are listed.

**Substances of very high concern**

|                               |                 |                      |                 |                 |         |
|-------------------------------|-----------------|----------------------|-----------------|-----------------|---------|
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|                               |                 |                      | (Germany)       |                 |         |

SECTION 15: Regulatory information

None of the components are listed.

Annex XVII - Restrictions on the manufacture, placing on the market and use of certain dangerous substances, mixtures and articles

| Product/ingredient name | %      | Designation [Usage] |
|-------------------------|--------|---------------------|
| Iloform RN 3017         | 95-100 | 3                   |

Labelling Not applicable.

Other regulations

REACH Status The company, as identified in Section 1, sells this product in the EU in compliance with the current requirements of REACH.

United States inventory (TSCA 8b) Not determined.

Australia inventory (AIIIC) All components are listed or exempted.

Canada inventory At least one component is not listed in DSL but all such components are listed in NDSL.

China inventory (IECSC) All components are listed or exempted.

Japan inventory (CSCL) At least one component is not listed.

Korea inventory (KECI) At least one component is not listed.

Philippines inventory (PICCS) At least one component is not listed.

Taiwan Chemical Substances Inventory (TCSI) Not determined.

Explosive precursors Not applicable.

Ozone depleting substances (EU 2024/590)

Not listed.

Prior Informed Consent (PIC) (649/2012/EU)

Not listed.

Persistent Organic Pollutants

Not listed.

EU - Water framework directive - Priority substances

None of the components are listed.

Seveso Directive

This product is not controlled under the Seveso Directive.

National regulations

Hazardous incident ordinance

This product is not controlled under the Germany Hazardous Incident Ordinance.

Hazard class for water 1 (classified according AwsV)

Prohibited Chemicals Regulation (ChemVerbotsV) When placed on the market in Germany, this product is not subject to the Prohibited Chemicals Regulation (ChemVerbotsV).

Occupational restrictions Observe employment restrictions in the following:  
Gesetz zum Schutz der arbeitenden Jugend (Jugendarbeitsschutzgesetz – JArbSchG)  
Gesetz zum Schutz von Müttern bei der Arbeit, in der Ausbildung und im Studium (Mutterschutzgesetz – MuSchG)

15.2 Chemical safety assessment A Chemical Safety Assessment has been carried out for one or more of the substances within this mixture. A Chemical Safety Assessment has not been carried out for the mixture itself.

SECTION 16: Other information

|                            |                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Abbreviations and acronyms | ADN = European Provisions concerning the International Carriage of Dangerous Goods by Inland Waterway<br>ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road<br>ATE = Acute Toxicity Estimate<br>BCF = Bioconcentration Factor<br>CAS = Chemical Abstracts Service<br>CLP = Classification, Labelling and Packaging Regulation [Regulation (EC) No. 1272/2008]<br>CSA = Chemical Safety Assessment |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                        |                 |               |                 |          |           |
|------------------------|-----------------|---------------|-----------------|----------|-----------|
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SECTION 16: Other information

CSR = Chemical Safety Report  
DMEL = Derived Minimal Effect Level  
DNEL = Derived No Effect Level  
EINECS = European Inventory of Existing Commercial chemical Substances  
ES = Exposure Scenario  
EUH statement = CLP-specific Hazard statement  
EWC = European Waste Catalogue  
GHS = Globally Harmonized System of Classification and Labelling of Chemicals  
IATA = International Air Transport Association  
IBC = Intermediate Bulk Container  
IMDG = International Maritime Dangerous Goods  
LogPow = logarithm of the octanol/water partition coefficient  
MARPOL = International Convention for the Prevention of Pollution From Ships, 1973 as modified by the Protocol of 1978. ("Marpol" = marine pollution)  
OECD = Organisation for Economic Co-operation and Development  
PBT = Persistent, Bioaccumulative and Toxic  
PNEC = Predicted No Effect Concentration  
REACH = Registration, Evaluation, Authorisation and Restriction of Chemicals Regulation [Regulation (EC) No. 1907/2006]  
RID = The Regulations concerning the International Carriage of Dangerous Goods by Rail  
RRN = REACH Registration Number  
SADT = Self-Accelerating Decomposition Temperature  
SVHC = Substances of Very High Concern  
STOT-RE = Specific Target Organ Toxicity - Repeated Exposure  
STOT-SE = Specific Target Organ Toxicity - Single Exposure  
TWA = Time weighted average  
UN = United Nations  
UVCB = Complex hydrocarbon substance  
VOC = Volatile Organic Compound  
vPvB = Very Persistent and Very Bioaccumulative  
Varies = may contain one or more of the following 64741-88-4 / RRN 01-2119488706-23, 64741-89-5 / RRN 01-2119487067-30, 64741-95-3 / RRN 01-2119487081-40, 64741-96-4/ RRN 01-2119483621-38, 64742-01-4 / RRN 01-2119488707-21, 64742-44-5 / RRN 01-2119985177-24, 64742-45-6, 64742-52-5 / RRN 01-2119467170-45, 64742-53-6 / RRN 01-2119480375-34, 64742-54-7 / RRN 01-2119484627-25, 64742-55-8 / RRN 01-2119487077-29, 64742-56-9 / RRN 01-2119480132-48, 64742-57-0 / RRN 01-2119489287-22, 64742-58-1, 64742-62-7 / RRN 01-2119480472-38, 64742-63-8, 64742-65-0 / RRN 01-2119471299-27, 64742-70-7 / RRN 01-2119487080-42, 72623-85-9 / RRN 01-2119555262-43, 72623-86-0 / RRN 01-2119474878-16, 72623-87-1 / RRN 01-2119474889-13

Procedure used to derive the classification according to Regulation (EC) No. 1272/2008 [CLP/GHS]

| Classification          | Justification      |
|-------------------------|--------------------|
| Asp. Tox. 1, H304       | Calculation method |
| Aquatic Chronic 3, H412 | Calculation method |

Full text of abbreviated H statements

H302 Harmful if swallowed.  
H304 May be fatal if swallowed and enters airways.  
H315 Causes skin irritation.  
H318 Causes serious eye damage.  
H361d Suspected of damaging the unborn child.  
H400 Very toxic to aquatic life.  
H410 Very toxic to aquatic life with long lasting effects.  
H412 Harmful to aquatic life with long lasting effects.  
H413 May cause long lasting harmful effects to aquatic life.  
EUH066 Repeated exposure may cause skin dryness or cracking.

Full text of classifications [CLP/GHS]

Acute Tox. 4 ACUTE TOXICITY - Category 4  
Aquatic Acute 1 SHORT-TERM (ACUTE) AQUATIC HAZARD - Category 1  
Aquatic Chronic 1 LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 1  
Aquatic Chronic 3 LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 3  
Aquatic Chronic 4 LONG-TERM (CHRONIC) AQUATIC HAZARD - Category 4  
Asp. Tox. 1 ASPIRATION HAZARD - Category 1  
Eye Dam. 1 SERIOUS EYE DAMAGE/EYE IRRITATION - Category 1  
Repr. 2 REPRODUCTIVE TOXICITY - Category 2  
Skin Irrit. 2 SKIN CORROSION/IRRITATION - Category 2

History

Date of issue/ Date of revision 29/10/2025.  
Date of previous issue 17/03/2025.  
Prepared by Product Stewardship

|                        |                 |               |                 |          |           |
|------------------------|-----------------|---------------|-----------------|----------|-----------|
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SECTION 16: Other information

Indicates information that has changed from previously issued version.

Notice to reader

All reasonably practicable steps have been taken to ensure this data sheet and the health, safety and environmental information contained in it is accurate as of the date specified below. No warranty or representation, express or implied is made as to the accuracy or completeness of the data and information in this data sheet.

The data and advice given apply when the product is sold for the stated application or applications. You should not use the product other than for the stated application or applications without seeking advice from BP Group.

It is the user's obligation to evaluate and use this product safely and to comply with all applicable laws and regulations. The BP Group shall not be responsible for any damage or injury resulting from use, other than the stated product use of the material, from any failure to adhere to recommendations, or from any hazards inherent in the nature of the material. Purchasers of the product for supply to a third party for use at work, have a duty to take all necessary steps to ensure that any person handling or using the product is provided with the information in this sheet. Employers have a duty to tell employees and others who may be affected of any hazards described in this sheet and of any precautions that should be taken. You can contact the BP Group to ensure that this document is the most current available. Alteration of this document is strictly prohibited.

## Annex to the extended Safety Data Sheet (eSDS)

Industrial

### Identification of the substance or mixture

|                    |                 |
|--------------------|-----------------|
| Product definition | Mixture         |
| Code               | 467692-DE15     |
| Product name       | Iloform RN 3017 |

### Section 1: Title

|                                      |                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Short title of the exposure scenario | Use of lubricants in high energy open processes - Industrial                                                                                                                                                                                                                                                                                                                 |
| List of use descriptors              | <b>Identified use name:</b> Use of lubricants in high energy open processes-Industrial<br><b>Process Category:</b> PROC01, PROC02, PROC08b, PROC17<br><b>Sector of end use:</b> SU03<br><b>Subsequent service life relevant for that use:</b> No.<br><b>Environmental Release Category:</b> ERC04<br><b>Specific Environmental Release Category:</b> ATIEL-ATC SPERC 4.Fi.v1 |

|                                                           |                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Processes and activities covered by the exposure scenario | Covers use of lubricants in high energy open processes, e.g. In high speed machinery such as metal rolling/forming or metal working fluids for machining and grinding. Includes associated product storage, material transfers, sampling and maintenance activities. |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Section 2 Operational conditions and risk management measures

### Section 2.1 Control of worker exposure

No exposure scenario is presented because the product is only classified as H304, EUH066 and/or H302 (Aspiration hazard – Category 1, Repeated exposure may cause skin dryness or cracking and/or Acute Toxicity (oral) – Category 4).

### Contributing scenarios: Operational conditions and risk management measures

### Section 2.2: Control of environmental exposure

#### Amounts used:

|                                                    |                      |
|----------------------------------------------------|----------------------|
| EU tonnage of risk determining substance per year: | 2.05E+02 Tonnes/year |
|----------------------------------------------------|----------------------|

#### Frequency and duration of use:

|               |     |
|---------------|-----|
| Emission days | 300 |
|---------------|-----|

#### Environment factors not influenced by risk management:

|                                    |     |
|------------------------------------|-----|
| Local freshwater dilution factor   | 10  |
| Local marine water dilution factor | 100 |

**Other operational conditions of use affecting environmental exposure:** Water-based (oil in water emulsion) or straight oil (contains no water) process

|                                                     |          |
|-----------------------------------------------------|----------|
| Release fraction to air (after typical onsite RMMs) | 1.00E-04 |
|-----------------------------------------------------|----------|

|                                                                   |   |
|-------------------------------------------------------------------|---|
| Release fraction to soil from process (after typical onsite RMMs) | 0 |
|-------------------------------------------------------------------|---|

|                                                                                                          |          |
|----------------------------------------------------------------------------------------------------------|----------|
| Release fraction to wastewater from process (after typical onsite RMMs and before sewage treatment plan) | 2.00E-11 |
|----------------------------------------------------------------------------------------------------------|----------|

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Use of lubricants in high energy open processes - Industrial

|                                                                                                                               |                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Technical conditions and measures at process level (source) to prevent release:</b>                                        | Common practices vary across sites thus conservative process release estimates used.                                                                                                                              |
| <b>Technical on-site conditions and measures to reduce or limit discharges, air emissions and releases to soil:</b>           | Prevent discharge of undissolved substance to or recover from onsite wastewater.<br>User sites are assumed to be provided with oil/water separators and waste water to be discharged via a sewage treatment plant |
| <b>Organisational measures to prevent/limit release from site:</b>                                                            | Do not apply industrial sludge to natural soils.<br>Sludge should be incinerated, contained or reclaimed.                                                                                                         |
| <b>Conditions and measures related to municipal sewage treatment plant:</b>                                                   |                                                                                                                                                                                                                   |
| Estimated substance removal from wastewater via on-site sewage treatment                                                      | 69                                                                                                                                                                                                                |
| Assumed domestic sewage treatment plant flow rate (m3/d)                                                                      | 2.00E+3                                                                                                                                                                                                           |
| Maximum allowable site tonnage (M <sub>Safe</sub> ) based on release following total wastewater treatment removal as product: | 797.3                                                                                                                                                                                                             |
| <b>Conditions and measures related to external treatment of waste for disposal:</b>                                           | External treatment and disposal of waste should comply with applicable local and/or national regulations.                                                                                                         |
| <b>Conditions and measures related to external recovery of waste:</b>                                                         | External recovery and recycling of waste should comply with applicable local and/or national regulations.                                                                                                         |

### Section 3: Exposure estimation and reference to its source

|                                                                      |                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure estimation and reference to its source - Environment</b> |                                                                                                                                                                                                                                        |
| Exposure assessment (environment):                                   | Used ECETOC TRA model (May 2010 release).                                                                                                                                                                                              |
| <b>Exposure estimation and reference to its source - Workers</b>     |                                                                                                                                                                                                                                        |
| Exposure assessment (human):                                         | No exposure scenario is presented because the product is only classified as H304, EUH066 and/or H302 (Aspiration hazard – Category 1, Repeated exposure may cause skin dryness or cracking and/or Acute Toxicity (oral) – Category 4). |

### Section 4: Guidance to check compliance with the exposure scenario

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Environment</b> | Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. Further details on scaling and control technologies are provided in SPERC factsheet. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required. For further information see <a href="http://www.ATIEL.org/REACH_GES">www.ATIEL.org/REACH_GES</a> |
| <b>Health</b>      | No exposure scenario is presented because the product is only classified as H304, EUH066 and/or H302 (Aspiration hazard – Category 1, Repeated exposure may cause skin dryness or cracking and/or Acute Toxicity (oral) – Category 4).                                                                                                                                                                                                                                                                                   |

## Annex to the extended Safety Data Sheet (eSDS)

Professional

### Identification of the substance or mixture

|                    |                 |
|--------------------|-----------------|
| Product definition | Mixture         |
| Code               | 467692-DE15     |
| Product name       | Iloform RN 3017 |

### Section 1: Title

|                                      |                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Short title of the exposure scenario | Use of lubricants in high energy open processes - Professional                                                                                                                                                                                                                                                                                                                  |
| List of use descriptors              | <b>Identified use name:</b> Use of lubricants in high energy open processes-Professional<br><b>Process Category:</b> PROC01, PROC02, PROC08a, PROC17<br><b>Sector of end use:</b> SU22<br><b>Subsequent service life relevant for that use:</b> No.<br><b>Environmental Release Category:</b> ERC08a<br><b>Specific Environmental Release Category:</b> ATIEL-ATC SPERC 8.Fp.v1 |

|                                                           |                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Processes and activities covered by the exposure scenario | Covers use of lubricants in high energy open processes, e.g. In high speed machinery such as metal rolling/forming or metal working fluids for machining and grinding. Includes associated product storage, material transfers, sampling and maintenance activities. |
|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Section 2 Operational conditions and risk management measures

#### Section 2.1 Control of worker exposure

No exposure scenario is presented because the product is only classified as H304, EUH066 and/or H302 (Aspiration hazard – Category 1, Repeated exposure may cause skin dryness or cracking and/or Acute Toxicity (oral) – Category 4).

#### Contributing scenarios: Operational conditions and risk management measures

#### Section 2.2: Control of environmental exposure

##### Amounts used:

|                                                    |                      |
|----------------------------------------------------|----------------------|
| EU tonnage of risk determining substance per year: | 2.05E+02 Tonnes/year |
|----------------------------------------------------|----------------------|

##### Frequency and duration of use:

|               |     |
|---------------|-----|
| Emission days | 365 |
|---------------|-----|

##### Environment factors not influenced by risk management:

|                                    |     |
|------------------------------------|-----|
| Local freshwater dilution factor   | 10  |
| Local marine water dilution factor | 100 |

**Other operational conditions of use affecting environmental exposure:** Negligible wastewater emissions as process operates without water contact.

|                                                                                                          |          |
|----------------------------------------------------------------------------------------------------------|----------|
| Release fraction to air (after typical onsite RMMs)                                                      | 1.00E-04 |
| Release fraction to soil from process (after typical onsite RMMs)                                        | 1E-03    |
| Release fraction to wastewater from process (after typical onsite RMMs and before sewage treatment plan) | 1.00E-03 |

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Use of lubricants in high energy open processes - Professional

|                                                                                                                               |                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Technical conditions and measures at process level (source) to prevent release:</b>                                        | Common practices vary across sites thus conservative process release estimates used.                      |
| <b>Technical on-site conditions and measures to reduce or limit discharges, air emissions and releases to soil:</b>           | Prevent discharge of undissolved substance to or recover from onsite wastewater.                          |
| <b>Organisational measures to prevent/limit release from site:</b>                                                            | Do not apply industrial sludge to natural soils.<br>Sludge should be incinerated, contained or reclaimed. |
| <b>Conditions and measures related to municipal sewage treatment plant:</b>                                                   |                                                                                                           |
| Estimated substance removal from wastewater via on-site sewage treatment                                                      | 69                                                                                                        |
| Assumed domestic sewage treatment plant flow rate (m3/d)                                                                      | 2.00E+3                                                                                                   |
| Maximum allowable site tonnage (M <sub>Safe</sub> ) based on release following total wastewater treatment removal as product: | 6.7                                                                                                       |
| <b>Conditions and measures related to external treatment of waste for disposal:</b>                                           | External treatment and disposal of waste should comply with applicable local and/or national regulations. |
| <b>Conditions and measures related to external recovery of waste:</b>                                                         | External recovery and recycling of waste should comply with applicable local and/or national regulations. |

### Section 3: Exposure estimation and reference to its source

|                                                                      |                                                                                                                                                                                                                                        |
|----------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure estimation and reference to its source - Environment</b> |                                                                                                                                                                                                                                        |
| <b>Exposure assessment (environment):</b>                            | Used ECETOC TRA model (May 2010 release).                                                                                                                                                                                              |
| <b>Exposure estimation and reference to its source - Workers</b>     |                                                                                                                                                                                                                                        |
| <b>Exposure assessment (human):</b>                                  | No exposure scenario is presented because the product is only classified as H304, EUH066 and/or H302 (Aspiration hazard – Category 1, Repeated exposure may cause skin dryness or cracking and/or Acute Toxicity (oral) – Category 4). |

### Section 4: Guidance to check compliance with the exposure scenario

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Environment</b> | Guidance is based on assumed operating conditions which may not be applicable to all sites; thus, scaling may be necessary to define appropriate site-specific risk management measures. Further details on scaling and control technologies are provided in SPERC factsheet. If scaling reveals a condition of unsafe use (i.e., RCRs > 1), additional RMMs or a site-specific chemical safety assessment is required. For further information see <a href="http://www.ATIEL.org/REACH_GES">www.ATIEL.org/REACH_GES</a> |
| <b>Health</b>      | No exposure scenario is presented because the product is only classified as H304, EUH066 and/or H302 (Aspiration hazard – Category 1, Repeated exposure may cause skin dryness or cracking and/or Acute Toxicity (oral) – Category 4).                                                                                                                                                                                                                                                                                   |